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where n is the number of moles of gas and V is the volume of the container

$$\Rightarrow P' \alpha \frac{n^2}{V^2}$$

$$\Rightarrow P' = a\frac{n^2}{V^2}$$

where a is proportionality constant and depends on the nature of gas

Therefore,

$$P_{ideal} = P + \frac{an^2}{V^2} \quad \text{------ (6.20)}$$

Volume Correction

As every individual molecule of a gas occupies a certain volume, the actual volume is less than the volume of the container, V. Van der Waals introduced a correction factor V' to this effect. Let us calculate the correction term by considering gas molecules as spheres.

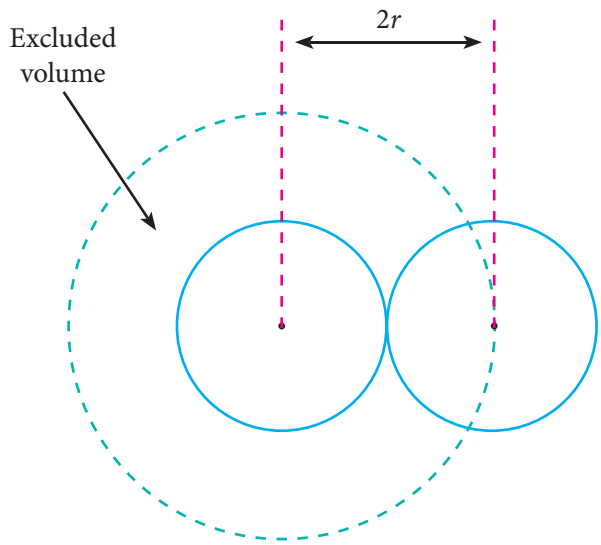


Figure. 6.11 Excluded volume

V = excluded volume

Excluded volume for two molecules

$$= \frac{4}{3}\pi(2r)^3$$

$$= 8\left(\frac{4}{3}\pi r^3\right) = 8\ V_m$$

where v_m is a volume of a single molecul e

Excluded volume for single molecule

$$= \frac{8\ V_m}{2} = 4\ V_m$$

Excluded volume for n molecule

$$= n\ (4\ V_m) = nb$$

Where b is van der waals constant whch is equal to $4\ V_m$

$$\Rightarrow V' = nb$$

$$V_{ideal} = V - nb \text{ ----- (6.21)}$$

Replacing the corrected pressure and volume in the ideal gas equation PV=nRT we get the van der Waals equation of state for real gases as below,

$$\left(P + \frac{an^2}{V^2}\right)(V - nb) = nRT \ . \text{----(6.22)}$$

The constants *a* and *b* are van der Waals constants and their values vary with the nature of the gas. It is an approximate formula for the non-ideal gas.

6.6 Pressure-Volume isotherms of Carbon dioxide

Andrew's isotherm

Thomas Andrew gave the first complete data on pressure-volume-temperature of a substance in the gaseous and liquid states. He plotted isotherms of carbon dioxide at different temperatures which is shown in Figure. 6.12. From the plots we can infer the following.